

CONTACT

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📍 Brisbane, QLD, Australia
Available: Jan 2027

TECHNOLOGIES

Languages

Python, SQL, Java, C, Haskell,
Dafny, Scala, MATLAB,
Shell/Bash, L^AT_EX

Machine Learning & AI

PyTorch, HuggingFace, Scikit-learn,
PyTorch Lightning, Faiss, NLTK

Data Processing & Databases

NumPy, Pandas, Spark, HDFS,
MongoDB, Redis, Relational DBs

Cloud & Infrastructure

Docker, Kubernetes (K8s), AWS,
GCP, OCI, S3, GCS, Slurm / HPC

RESEARCH INTERESTS

Machine Learning
Recommender Systems
Natural Language Processing
Large Language Models
Graph Neural Networks
Embedding Optimization
Model Scalability Optimization
Cloud & Distributed Systems
Big Data Processing

XURONG (JASON) LIANG

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Ph.D. Candidate
Computer Science

SUMMARY

Ph.D. candidate in Computer Science at the University of Queensland, specializing in efficient recommender systems, lightweight embeddings, graph collaborative filtering, and LLM-based recommendation. Published in SIGIR, RecSys, TOIS, ICDM, and KDD, with research spanning scalable recommendation, embedding compression, and model efficiency. Latest Ph.D. work investigates efficient compact LLMs for recommendation. Thesis submission expected in September 2026, oral defense anticipated in late 2026, and full-time availability from January 2027.

EDUCATION

Doctor of Philosophy (Ph.D.) in Computer Science
University of Queensland, Australia

2023 — Late 2026

Ph.D. project ***Toward Efficient Lightweight Recommender Systems***

Advised by Prof. Hongzhi Yin and Assoc. Prof. Rocky Tong Chen.

Thesis submission expected: September 2026; oral defense anticipated: late 2026; available full-time: January 2027.

Bachelor of Computer Science (Honors)
University of Queensland, Australia

2022

Honors project ***PEP++: Improved Pruning-Based Embedding Dimension Optimization for Recommender Systems***

Advised by Assoc. Prof. Rocky Tong Chen.

Awarded an **Honors Class I Degree (cumulative GPA 7.000/7.000)**.

Bachelor of Computer Science
University of Queensland, Australia

2019 — 2021

Graduated with cumulative **GPA 6.792/7.000**. Major in **Data Science**.

High School Diploma
Indooroopilly State High School, Australia

2016 — 2018

Graduated with **OP4** (Highest: 1, Lowest: 25).

PUBLICATIONS

LANGUAGES

English

Bilingual

High proficiency in reading and writing

Chinese

Mandarin

Native Fluency

Chinese

Cantonese

Native Fluency

Empowering Compact LLMs with Fusion of Layer-wise Exits for Recommendation

Latest first-authored Ph.D. work on efficient LLM-based recommendation. Paper and code will be made available upon acceptance.

Xurong Liang, Tong Chen, Quoc Viet Hung Nguyen, Jianxin Li, Xiangliang Zhang, Hongzhi Yin

Under Review

In-context Learning for Addressing User Cold-start in Sequential Movie Recommenders

Xurong Liang, Vu Nguyen, Vuong Le, Paul Albert, Julien Monteil

RecSys '25

Lightweight Embeddings with Graph Rewiring for Collaborative Filtering

Xurong Liang, Tong Chen, Wei Yuan, and Hongzhi Yin

TOIS '25

Lightweight Embeddings for Graph Collaborative Filtering

Xurong Liang, Tong Chen, Lizhen Cui, Yang Wang, Meng Wang, and Hongzhi Yin

SIGIR '24

Hate Speech Detection with Generalizable Target-aware Fairness

Tong Chen, Danny Wang, **Xurong Liang**, Marten Risius, Gianluca Demartini and Hongzhi Yin

KDD '24

Learning Compact Compositional Embeddings via Regularized Pruning for Recommendation

Xurong Liang, Tong Chen, Quoc Viet Hung Nguyen, Jianxin Li, and Hongzhi Yin

ICDM '23

ACHIEVEMENTS

Dean's Commendation for Academic Excellence — Bachelor of Computer Science (Hons)

Awarded to students who obtained a grade point average of 6.60 or higher in a single semester.

2022

Dean's Commendation for Academic Excellence — Bachelor of Computer Science

Awarded to students who obtained a grade point average of 6.60 or higher in a single semester.

2019 — Jul 2021

Cloud Computing INFS3208 Best Project Runner-up Award University of Queensland, Australia Awarded for outstanding performance in INFS3208 Cloud Computing individual project competition 2021.	2021
Prentice Scholar 2020 University of Queensland, Australia Prentice Scholars are students whose academic performance sits within the top five percent of the undergraduate Computer Science and Information Technology cohort at The University of Queensland.	2020
Subject Prize Award – Information Processing and Technology Indooroopilly State High School, Australia Ranked 1st in the Information Processing and Technology course cohort.	2017

SCHOLARSHIPS

Australian Government Research Training Program (RTP) Scholarship	2023 — 2026
Google Research Travel Scholarship	2024
UQ International Onshore Merit Scholarship	2022
Winter Research Scholarship	2021

ACADEMIC & RESEARCH EXPERIENCE

Ph.D. Researcher University of Queensland, Australia - Designed and optimized lightweight recommender-system architectures leveraging Graph Neural Networks (GNNs), Large Language Models (LLMs), and Mixture-of-Experts (MoE). - Developed efficient algorithmic frameworks to reduce embedding storage footprint, memory consumption, and computational overhead for large-scale recommendation. - Orchestrated model training and data-processing pipelines on high-performance computing clusters using Python, PyTorch, and Slurm. - Published first-author research in top-tier venues including SIGIR, RecSys, TOIS, and ICDM.	Jan 2023 — Present
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Applied Scientist Intern

Oct 2024 — May 2025

Amazon, Australia

- Developed an **LLM-based in-context learning** framework for user cold-start recommendation in a commercial video streaming platform; resulting work was published at **RecSys '25**.
- Built an **AWS-based ML/data pipeline** using **EMR, SageMaker Studio, Bedrock**, and **S3** to process large-scale user interaction data, perform distributed preprocessing and analysis, and support downstream ML modeling.
- Collaborated with scientists and engineers under the guidance of Dr. Julien Monteil, Dr. Vu Nguyen, Dr. Vuong Le, and Dr. Paul Albert.

Head Tutor & Teaching Assistant

Feb 2022 — Jul 2026

University of Queensland, Australia

- Served as Head Tutor for **INFS7901 - Database Principles** and **INFS3208/INFS7208 - Cloud Computing**.
- Delivered practical and tutorial sessions, managed course administration, and coordinated assignment and examination marking.
- Translated cloud, database, and distributed-systems concepts into accessible teaching materials for students with diverse technical backgrounds.
- Mentored student projects using **Docker, Docker Swarm, K8s, SQL, MongoDB, Redis, Faiss, Spark, HDFS, Amazon S3**, and **Google Cloud Storage**.

Casual Research Assistant

Sep 2021 — Sep 2024

University of Queensland, Australia

- Conducted research experiments, data preprocessing, baseline testing, and algorithm implementation for **Explainable Recommender Systems** and **Hate Speech Detection**.
- Contributed to a novel hate speech detection framework successfully published at **KDD '24**.

Winter Research Scholar

Jun 2021 — Jul 2021

University of Queensland, Australia

- Conducted the project **Dynamic Topic Modelling on Social Media**, using **Natural Language Processing (NLP)** to model topic changes on social media during the first 100 days of COVID-19.
- Produced project outputs under the supervision of Prof. Zi (Helen) Huang and Assoc. Prof. Rocky Tong Chen.
- Published project visualization at <https://topicvis.nifu.me/>.

Voluntary Teaching Assistant

Dec 2019 — Jan 2020

Dalu Primary School (大芦小学), Yulin, Guangxi 537000, China

- Taught English and Information Technology courses to elementary school students.
- Organized homework materials and supported marking activities.

ADDITIONAL EXPERIENCE

Self-employed Courier Delivery Service

Dec 2020 — Feb 2021

Brisbane, QLD 4000, Australia

- Contracted last-mile delivery orders with courier providers including 1stEx[®] and CouriersPlease[®] across the Greater Brisbane Region.

Expo Helper**Jul 2017****AAS Education & Student Services, Spring Hill, QLD
4000, Australia**

- Assisted prospective students with inquiries regarding further study abroad.
- Supported student recruitment and encouraged prospective genuine students to apply on-site.

Voluntary International Student Ambassador**Jan 2017****Indooroopilly State High School, Australia**

- Assisted new students during International Student Orientation.
- Provided school tours for visiting delegates.

REFEREES

Assoc. Prof. Rocky Tong Chen**University of Queensland, Australia**

Ph.D. Principal Advisor

Email: tong.chen@uq.edu.au

Prof. Hongzhi Yin**University of Queensland, Australia**

Ph.D. Co-Advisor

Email: h.yin1@uq.edu.au

Assoc. Prof. Sen Wang**University of Queensland, Australia**

Course Coordinator of Cloud Computing

Email: sen.wang@uq.edu.au

Dr. Julien Monteil**Amazon, Australia**

Applied Scientist Internship Manager

Email: jul@amazon.com